

01 | Whole system: what and why

5 October 2026 | Architecture d7c0fd22c plus current diagram-review corrections | Frozen PRD: 2 October 2026 | Draft design, not executed system acceptance.

PRODUCTION / FIXED EIGHT-STAGE PATH



Every transition: committed evidence + Gate decision + release. Final report -> publication.

- **Goal:** a local question-to-report research workflow with durable evidence. Exact shared contracts let different builders connect modules without inventing product behavior. Internal algorithms remain implementation choices.
- **Deployment:** one Dockerized modular monolith, with protected internal processes where trust requires isolation. This keeps deployment simple while separating credentials, generated code and hidden fixtures. **Pattern:** [Docker restricted containers](#); process separation is our local design.
- **Three tracks:** fixed production research; required offline RSI; isolated PRD-whitelisted Phase 2 experiments. Planner experiments never rewrite production. Spec Kit, live replanning, fusion and unrestricted delegation are outside this architecture work.
- **Inventory:** eight research work CCs + one shared verifier + three search operators. Planner, Gate host, store, source extraction, arithmetic and publication are ordinary modules. **Why:** independent governance/reuse needs a capsule; mechanical helpers do not.
- **Contracts:** capability-named capsules compose through exact versioned ports and pinned resources. **Pattern:** [Kubeflow component specifications](#), borrowed for interfaces rather than its runtime.
- **Authority:** frozen PRD sets scope; owner inputs supply domain detail; architecture chooses shared interfaces. One owner per schema/API/record, with linked generated views. **Pattern:** [Nygard ADRs](#) preserve rationale and supersession.
- **Current state:** proposed modules and concrete contracts are documented. Architecture remains draft; document checks and AI review do not prove a working implementation, platform safety or approval.

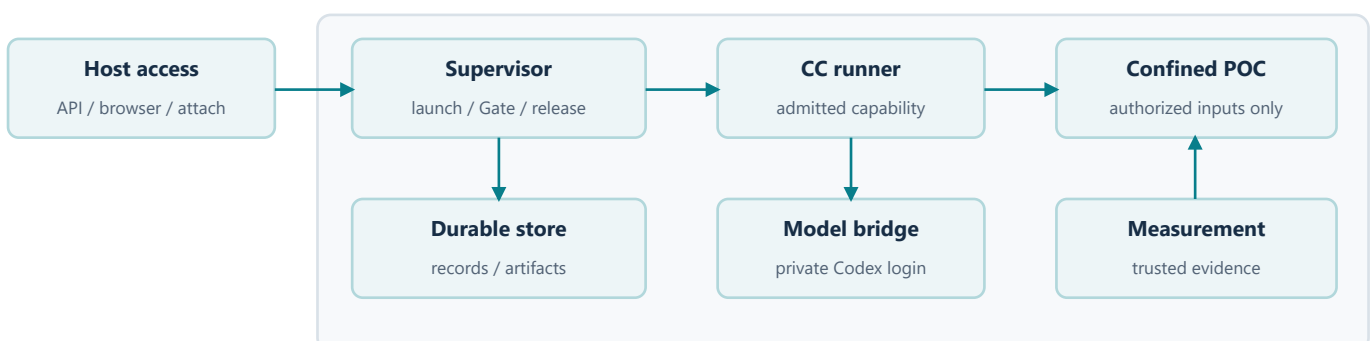
Canonical owners: [README.md](#); [authority.md](#); [system/overall-draft.md](#); [system/modules.md](#); [m1/pipeline.md](#); [policies.md](#)

02 | Connections, control and evidence

- **Entry -> supervisor:** CLI, local Web/TUI and headless HTTP use shared run authority. **Why:** multiple clients must observe one execution state.
- **Supervisor -> runner:** a managed subprocess executes admitted versions through authenticated frames; the supervisor keeps its client. **Runner -> broker -> model bridge:** the protected model adapter mediates Codex turns.
- **Runtime order:** validate inputs -> freeze bindings -> run CC -> store Artifact/Observation -> evaluate Gate -> store Verification -> store release -> dispatch successor. **Pattern:** Temporal durable execution; our records, not Temporal's runtime, authorize progress.
- **Gate:** the Gate host folds deterministic checks and applicable shared-verifier assessment under pinned stage criteria. **Why:** one policy interface avoids duplicated stage hosts. **Pattern:** OPA policy/data separation.
- **Persistence:** the trusted store commits public records and content-addressed artifacts; required capture feeds Data Foundation views. Atomic/recoverable publication and frozen hashes prevent partial or mutable evidence becoming authority. Durability still needs crash/fsync tests.
- **Failure/recovery:** failed output, Verification or release write blocks the successor. Recovery reuses committed work/decisions; explicit re-execution of failed or interrupted work creates a new attempt. Changed inputs/settings require a new run.
- **Duplicates/quotas:** stable request hashes reuse results or conflict on changed bytes. Committed model reservations enforce bounded calls; uncertain effects are not silently retried. Native engine retry/cache behavior must be constrained by the CC adapter.
- **Reuse:** pinned OpenJiuwen/agent-core workflow, backend, local session, transport and views sit behind replaceable adapters. Source pin 6cc05c36b and agent-core 9e339019 are distinct from this architecture revision; reuse does not imply native security/durability compliance.

Canonical owners: [system/integration.md](#); [system/reuse-audit-2026-10-05.md](#); [capsule/toolchain.md](#); [capsule/runner.md](#); [capsule/gate-host.md](#); [system/storage.md](#); [system/lifecycle.md](#)

PRODUCTION / MODULE CONNECTIONS INSIDE ONE DEPLOYABLE



03 | First CCs: behavior and connections

- **Construction order:** foundation and minimal admitted fixture -> Requirement Compilation -> shared verifier/Gate -> durable continuation -> Search/operators/Screening. This is incremental construction, not execution order. Governed slices need functioning storage, admission, runner and release authority.
- **research.compile_brief (skill):** intake + source_text + optional deterministic intent_ir -> research_brief. Interprets requirements with source grounding. Gate: research.accept_brief.v1. Output connects to Search and downstream stages.
- **research.search_ideas (tool):** research_brief + intake -> idea_set. Collects bounded retained evidence through search operators. Gate: research.accept_ideas.v1. Output connects to Screening.
- **research.select_opportunity (tool):** idea_set + research_brief -> opportunity_card. Prompt-led assessment plus protected deterministic filtering/ranking. Gate: research.accept_card.v1. Output connects to Hypothesis.
- **Screening policy:** one consolidation pass; Novelty, Technical Feasibility and Compute Alignment score 1-5; sum ranks eligible candidates. Frozen dependency conflict/unknown is excluded; canonical source-idea ordering breaks ties. Preserve rejection reasons; no eligible card halts for human review.
- **research.form_hypothesis:** opportunity_card + research_brief + intake -> hypothesis_blueprint. Freezes claims, protocol, resources and predicates before code generation. Gate: research.accept_hypothesis.v1.
Pattern: [OSF preregistration](#).
- **Packaging/why:** Declaration, implementation/prompt and checks are pinned by capability version. Exact ports support independent builders. **Pattern:** [Kubeflow typed components](#). Capsules belong to the admitted library and execute under the runner; the architecture does not claim they are already implemented.

Canonical owners: [system/build-order.md](#); [m1/pipeline.md](#); [m1/capability-designs.md](#); [m1/requirement-capsule.md](#); [m1/search-capsule.md](#); [m1/screening.md](#); [m1/hypothesis.md](#)

04 | Remaining CCs and scientific path

- **research.build_poc:** Blueprint + Brief + intake -> poc_bundle. Gate: research.accept_poc.v1. Default: two generation calls, patch then harness, with deterministic packaging/environment checks; no third repair call.
- **research.run_benchmark:** poc_bundle + Blueprint -> benchmark_payload. Gate: research.accept_benchmark.v1. Coordinates frozen baseline/treatment execution; trusted measurement modules, not generated code, own measurement evidence.
- **research.evaluate_results:** benchmark_payload + Blueprint + Brief -> evaluation_verdict. Gate: research.accept_evaluation.v1. Frozen deterministic comparison plus one plausibility turn. **Pattern:** Python Decimal for pinned arithmetic.
- **Scientific outcomes:** PASS, FAIL, INCONCLUSIVE or preregistered CONDITIONALLY_ACCEPTABLE. Valid negative/middle-zone results can proceed to report; missing/corrupt evidence blocks infrastructure acceptance. **Why:** evidence validity and research success answer different questions.
- **research.write_report:** verdict + benchmark + Brief + ideas + opportunity + Blueprint + authorized StageContext -> research_report. Gate: research.accept_report.v1. One synthesis call; ordinary publisher commits the final manifest.
- **research.verifier:** shared semantic evidence/criteria assessment under pinned stage profiles. Gate host owns folding and Verification; the trusted supervisor owns release and dispatch. **Why:** common assessment infrastructure, distinct stage criteria.
- **Search operators:** op.scholarly_search, op.local_search and op.codesearch are independently governed retrieval capabilities serving declared callers. Search uses local/scholarly retrieval; code-location consumers can use CodeSearch. Nested declared retrieval produces retained evidence; bounded permissions mediate external/repository access. DeepSearch/code-search reuse remains behind adapters.
- **Optimization:** permitted prompt/helper changes preserve protected interfaces, ranking, policy and referee. Benchmark materials include EvalPlus, BEIR, CRAG and BIPIA as scoped candidates; assets need receipts/licenses and protected splits. No corpus acquisition or performance improvement is claimed.

Canonical owners: m1/pipeline.md; m1/capability-designs.md; m1/measurement-protocol.md; m1/evaluation.md; m1/delivery.md; m1/benchmarking-material.md

05 | Security, authentication and environment

- **Confinement:** generated-code processes use fixed identities, inner filesystem/PID/network namespaces, bounded syscalls and authenticated brokers. **Patterns:** [Bubblewrap namespaces](#) + [Docker seccomp](#). Their combination remains unvalidated.
- **Access:** admitted code and input snapshots are read-only; assigned attempt root is writable. Credentials, hidden fixtures, store, host home and other attempts are absent from child mounts/descriptors. No Docker socket or external workload network.
- **Lifecycle:** timeout/cancellation kills and reaps the owned namespace process tree before capture seals. **Why:** process-group signaling alone cannot prove that all generated grandchildren stopped.
- **Codex:** separate container login; persistent bridge-owned CODEX_HOME; one serialized owner and exclusive lock. No desktop credential copy or watcher. **Pattern:** [Codex account authentication](#) persistent-cache/refresh custody; real binary behavior still needs tests.
- **Replaceability:** ModelProvider/AuthProvider isolate endpoint and login changes. Production starts with Codex; alternative routing is approved/mockd isolated experimentation. Credentials never enter research prompts, configuration snapshots or benchmark export.
- **Environment:** one pinned Linux/Python 3.12 image targets Linux Engine/macOS Docker Desktop. Binary-only hashed offline wheelhouse. **Pattern:** [pip secure installs](#). Repository $\geq 3.11, < 3.14$ does not establish every wheel/GPU/platform combination.
- **Configuration:** defaults -> user -> project; security ceilings cannot be weakened. Runs pin settings; service changes require explicit idle restart. Doctor exercises the actual profile and fails closed with UNSUPPORTED_SECURITY_PROFILE when required boundaries are unavailable.

Canonical owners: [system/deployment.md](#); [system/environment.md](#); [system/model-auth.md](#); [capsule/process-boundary.md](#); [capsule/fixture-oracle.md](#); [open-issues.md](#)

06 | RSI, planner and benchmark interfaces

- **RSI path:** isolated proposal -> private trial/oracle -> paired comparisons -> sealed lineage/ablations -> final holdout -> Candidate -> ordinary admission -> admitted_inactive -> human activation.
- **Mutation/quotas:** freeze interfaces, referee, dependencies and splits. Loop limits: 30/session, 90/immutable-set lifetime; schedule 1 baseline + up to 23 proposals + up to five ablations. Final holdout is separate, once/session. Repeated-session final-aggregate adaptation remains a methodological risk.
- **Admission:** tested evidence or Puppet developer allowlisting; Puppet grants only exempt assurance after integrity checks. It cannot fabricate tests, certify, activate or issue runtime passes. **Pattern:** [MLflow version/alias separation](#), adapted to capsule availability/assurance/activation.
- **Planner:** a separate orchestration controller, default GPT-6.1 Sol through Codex. Isolated proposals select admitted versions/bindings. Deterministic validator checks types, closure, cycles, permissions, budgets, Gates and objective coverage; freeze requires committed validation authority.
- **Production distinction:** same plan representation, exact fixed template from committed intake; no model planning prerequisite. No live replanning, unrestricted delegation or autonomous repair. **Why:** stable baseline and replaceable experimental planning boundary.
- **Harness:** authenticated HTTP at host 127.0.0.1:8787 supports runs, status, abort, sealed exports and manifest-scoped downloads. One active run; duplicates reuse handles. A replaceable entry/export adapter connects Saurav's external harness. **Pattern:** [HTTP semantics](#) + [OpenAPI](#).
- **Export:** correlated identities, source/library/config pins, model calls, Gates/releases, timing, artifacts, scientific outcomes and failures. Missing telemetry is explicit. Saurav's pending schema changes the export adapter, not the research path. **Pattern:** [OpenTelemetry span/link correlation](#).

Canonical owners: [capsule/rsi-engine.md](#); [capsule/fixture-oracle.md](#); [capsule/admission.md](#); [capsule/library.md](#); [system/planner.md](#); [system/experiments.md](#); [system/benchmark-export.md](#)

07 | Adversarial review, evidence and handoff

- **Break cases:** lost Gate save, duplicate/uncertain paid call, credential/fixture read, raw-plan bypass, RSI quota reset, unauthorized export and scientific misclassification. Defenses are durable authority, reservations, private custody, deterministic validation and manifest checks; product fault injections remain unrun.
- **Fresh review corrections:** clarify incremental build order and committed-result recovery; distinguish execution kinds and exempt assurance. Fix stale verification API summaries and source-pin wording. No confirmed targeted owner-contract contradiction remains; this is bounded source review, not exhaustive assurance.
- **Highest residual risks:** exact Docker namespace/credential/process profile on each platform; native retry/cache adapters; fsync/crash recovery; independent semantic fixtures and trusted measurements; Codex refresh/cancellation; repeated-session holdout adaptation.
- **Recorded checks:** prior batches: 464 service checks, 20 source hashes, eight input pairs/18 joins, 131 payload checks, 136 story checks. Review-workflow batch: 1,782 links and 17 packet probes. Earlier diagrams: 50 blocks parsed, 12 selected renders. These are historical documentation evidence, not runtime results.
- **Quality policy:** cheap bounded packets and fresh failure/scope/coder roles; independent seam derivation; escalate shared authority/security disputes. **Patterns:** [Pact consumer/provider verification](#) + [Google small-change reviews](#). No measured token savings or unbreakability claim.
- **Coder handoff:** owners answer placement, API, order/recovery, persistence, security, environment and verification hooks. Start with authority/atlas -> modules -> capability designs -> handoff. Coders produce Spec Kit and actual test evidence; people receive concise decisions/risks.
- **Maintenance:** this report is a dated derived view. Owning pages, decisions and open obligations stay authoritative. An implementation or handoff release must pin current contracts and connected evidence; historical checkpoints cannot certify a later revision.

Canonical owners: [authority.md](#); [system/diagram-atlas.md](#); [system/coder-requirements.md](#); [system/handoff.md](#); [open-issues.md](#); [review-workflow.md](#); [reviews/2026-10-05-adversarial-design.md](#)

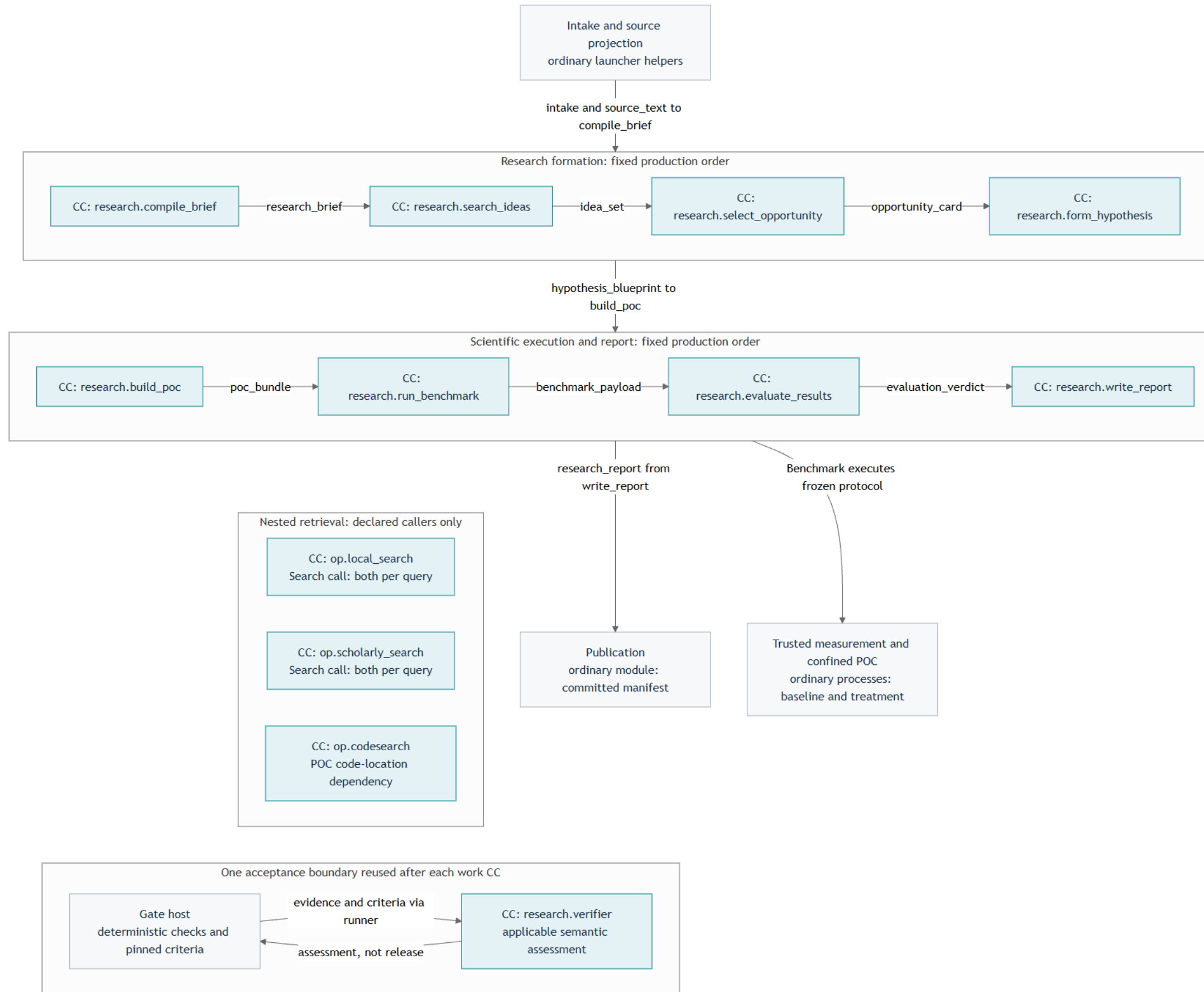
08 | Decisions, depth and development policy

- **The short answer:** one Dockerized local application; twelve capability identities; a fixed eight-step production research flow; protected Codex access; real durable Gates; offline RSI with human activation; isolated optional planning. These are design choices, not claims of implemented acceptance.
- **Why those choices:** one deployable reduces operational burden; typed CC contracts enable independent builders; shared Gate profiles avoid duplicate judgment infrastructure; persisted release authority stops unsafe continuation; provider/export adapters limit replacement cost. Pattern sources appear beside each design choice earlier in this report.
- **Depth 1 - explain the system:** the overview Mermaid views show every CC, nested helpers, track selection, trust placement and admission/activation. Dashed CONDITIONAL nodes may be absent from a particular run. They are not permission to omit required production stages or mandatory Search dependencies.
- **Depth 2 - define the boundaries:** the module map and capsule guides link single owning schemas, API/effect/error semantics, configuration, immutable references and write authority. The generated information-flow variants include all 23 required production bindings plus publication fan-in. Appendix maps show the core wiring and failure/recovery; full/filtered Mermaid views are linked from the sharing index.
- **Depth 3 - test the joins:** blind producer/consumer derivations and seam canaries; valid/missing/unknown-field fixtures; wrong versions/references; duplicates, timeout/cancellation, lost writes, restart, planner rejection, unauthorized access and scientific-negative completion. **Pattern:** [Pact boundary verification](#).
- **Depth 4 - prove the system:** independent component invocation with fake adjacent services, then integrated production, halt/recovery, RSI/admission, planner/dispatch and benchmark-export journeys. Observe actual records and successor invocation counts. Planned scenarios are not recorded passes.
- **Change policy:** update the owning contract and impacted consumers, profiles, records, diagrams/examples and coverage together; recheck affected evidence. Frozen sources stay verbatim. A released interface change requires explicit compatibility/migration, not silent edits to builders' assumptions.
- **How this helps coding:** separate owners build against agreed types and mockable public APIs, then connect without deciding product behavior. Coders translate the architecture into Spec Kit, choose internal algorithms and implement tests. Parallel work starts at stable shared boundaries; runtime safety still needs executed evidence.

Canonical owners: [presentation/showcase/README.md](#); [system/overall-draft.md](#); [system/information-flow.md](#); [authority.md](#); [system/modules.md](#); [m1/capability-designs.md](#); [system/coder-requirements.md](#); [system/verification.md](#); [review-workflow.md](#); [system/handoff.md](#)

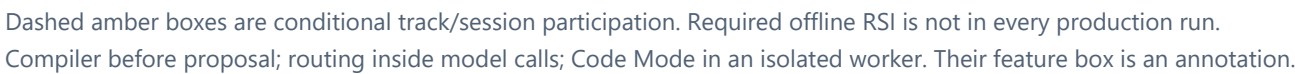
Every capsule in the research path

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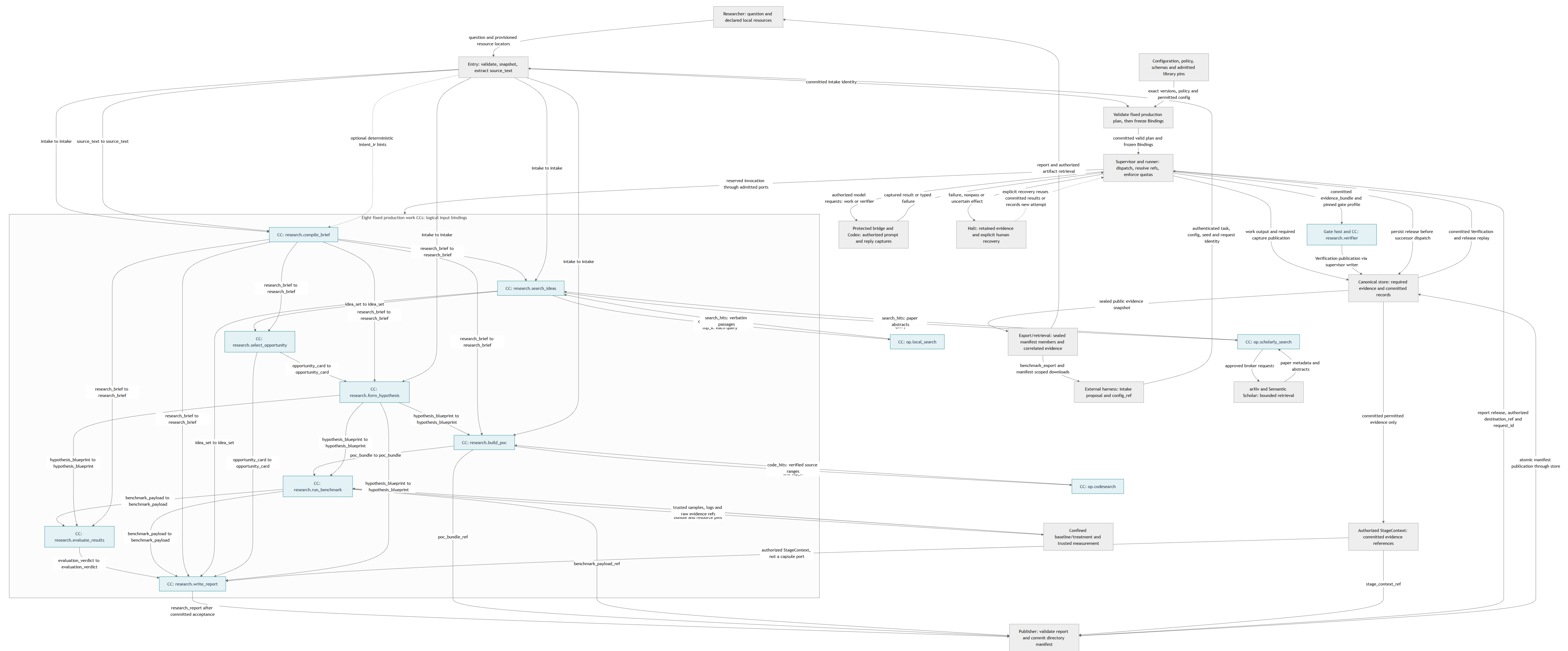
Eight fixed work CCs; three nested retrieval CCs; one shared verifier. Supporting insets are not later stages.
Search calls BOTH local and scholarly operators per query; POC calls CodeSearch. Gates apply after every work CC.

05 OCT 2026 / MERMAID SOURCE VIEW / DRAFT, NOT RUNTIME ACCEPTANCE



Detailed core: actual information bindings

05 OCT 2026 / MERMAID SOURCE VIEW / DRAFT, NOT RUNTIME ACCEPTANCE

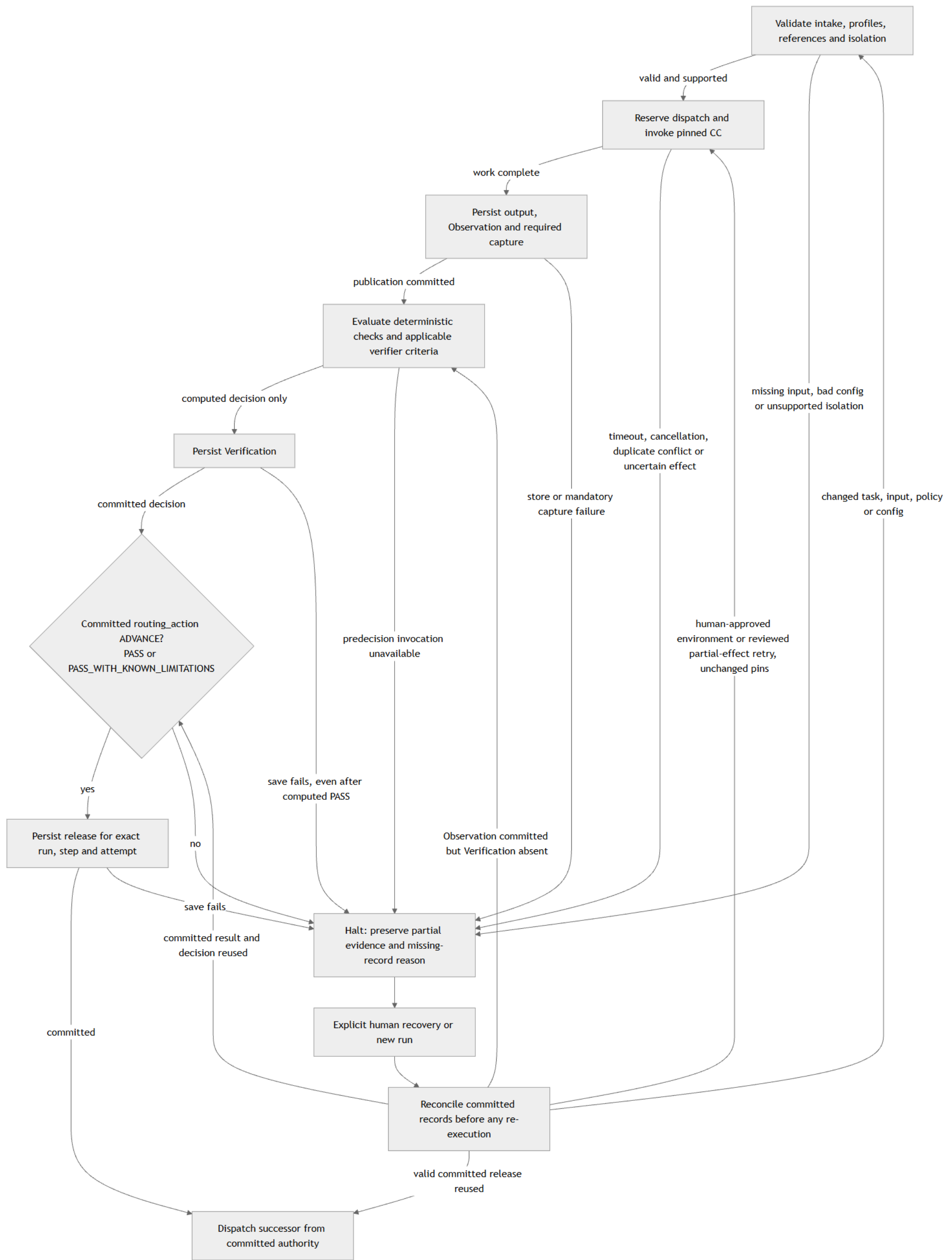


Zoom map: all 23 required production input bindings plus nested calls, publisher fan-in, I/O and durable authority.
Derived observability and offline RSI branches are hidden here, not disabled. Full and other filters are linked from the sharing index.

Editable Mermaid and canonical owner links: [system/information-flow.md](#)

Failure and recovery: what permits the next step?

05 OCT 2026 / MERMAID SOURCE VIEW / DRAFT, NOT RUNTIME ACCEPTANCE



Advancement requires committed PASS or PASS_WITH_KNOWN_LIMITATIONS and routing_action ADVANCE, then durable release. Ordinary failed checks produce Verification. Explicit recovery reuses committed evidence and never silently repeats uncertain work.